

EXERCISE SHEET 1

This is not the official exercise sheet from the course.

More difficult exercises are denoted by ★

Exercise 1.

Let the universe be the rational numbers $\mathbb{Q}$.

We define the following predicates

$$P(x) = \begin{cases} 1, & x \in \mathbb{Z} \\ 0, & \text{Else} \end{cases} \quad Q(x) = \begin{cases} 1, & x > 2 \\ 0, & \text{Else} \end{cases} \quad R(x, y, z) = \begin{cases} 1, & x^3 + y^3 = z^3 \\ 0, & \text{Else} \end{cases}$$

In this context, describe the following statements in words. Decide for each statements if its true or false.

1. $\exists x(P(x) \wedge Q(x))$
2. $\exists x(Q(x) \wedge \neg P(x))$
3. $\forall x(Q(x) \rightarrow P(x))$
4. $\forall x \forall y \exists z(R(x, y, z))$
5. $\exists x \exists y \exists z(P(x) \wedge P(y) \wedge R(x, y, z))$
6. ★ $\neg \exists x \exists y \exists z(P(x) \wedge P(y) \wedge P(z) \wedge R(x, y, z))$
7. ★ $\forall x \forall y \forall z((P(x) \wedge P(y) \wedge P(z) \wedge R(x, y, z)) \rightarrow (x = 0 \vee y = 0 \vee z = 0))$

Hint: For 6 and 7, look up Fermat's Last Theorem.

Solution 1.

1.

"There exists an integer greater than 2."

The statement is true

2.

"There exists a rational number that is greater than 2 and not an integer."

The statement is true

3.

"If a rational number is greater than 2, then it is an integer."

The statement is false

4.

"For all rational numbers x, y there exists a rational number z such that $x^3 + y^3 = z^3$."

The statement is false. Take $x = y = 1 \in \mathbb{Q}$. Then $x^3 + y^3 = 2$, which implies that $z = \sqrt[3]{2}$. But $\sqrt[3]{2} \notin \mathbb{Q}$

5.

"There exist two integers x, y and one rational number z , such that $x^3 + y^3 = z^3$."

The statement is true. Take $x = 0, y = 1, z = 1$.

6.

"There do not exist integers x, y, z such that $x^3 + y^3 = z^3$."

The statement is false. Take the trivial solution $x = y = z = 0$

7.

"For all integers x, y, z , if $x^3 + y^3 = z^3$, then at least one of them must be 0."

The statement is true. By Fermat's Last Theorem, $x^3 + y^3 = z^3$ has no non-zero integer solutions. Therefore, any integer solution must contain at least one variable equal to 0.

Exercise 2.

Let the universe be the integers $\mathbb{Z}$. Describe the following mathematical statements using predicate logic. Define your own suitable predicates and decide if the statement is true.

1. For every integer x , there exists a number y such that $x + y = 0$
2. There exists no largest natural number.
3. The sum of two odd numbers is even.
4. The product of any two numbers is strictly positive.
5. Any prime bigger than 5, is of the form $6n + 1$.
6. There exists an integer x such that for every integer y , $x \leq y$.
7. Every positive integer has an integer square root.
8. Every even integer can be expressed as the difference of two odd integers.
9. Every positive integer greater than 1 can be written as the sum of two prime numbers.
10. ★ There are infinitely many perfect squares.

Solution 2.

1.

The statement is true and can be written as

$$\forall x \exists y (x + y = 0)$$

2.

Let $N(x) = 1$ if and only if $x \in \mathbb{N}$. The statement is true and can be written as

$$\neg \exists x (N(x) \wedge \forall y (N(y) \rightarrow y \leq x))$$

3.

Let $O(x) = 1$ if and only if x is odd. Let $E(x) = 1$ if and only if x is even.

Then the statement is true and can be written as

$$\forall x \forall y ((O(x) \wedge O(y)) \rightarrow E(x + y))$$

4.

The statement is false and can be written as

$$\forall x \forall y (x \cdot y > 0)$$

5.

Let $P(x) = 1$ if and only if x is prime. This is false and can be written as

$$\forall x ((P(x) \wedge x > 5) \rightarrow \exists y (x = 6y + 1))$$

Counterexample: Take $p = 11$, which can't be written as $6n + 1$.

Note: it would be true if it were $6n \pm 1$

6.

This is false and can be written as

$$\exists x \forall y (x \leq y)$$

7.

This is false and can be written as

$$\forall x (x > 0 \rightarrow \exists y (y^2 = x))$$

Counterexample: $x = 2$ has no square root in $\mathbb{Z}$.

8.

Let $O(x) = 1$ if and only if x is odd. Let $E(x) = 1$ if and only if x is even.

The statement is true and can be written as

$$\forall x(E(x) \rightarrow \exists y \exists z(O(y) \wedge O(z) \wedge x = y - z))$$

For any x , take $y = x + 1$ and $z = 1$, both are odd and $y - z = x$.

9.

Let $P(x) = 1$ if and only if x is prime.

The statement is false and can be written as

$$\forall x(x > 1 \rightarrow \exists y \exists z(P(y) \wedge P(z) \wedge x = y + z))$$

Counterexample. Take $x = 11$. Then one of y, z must be even, so assume without loss of generality that $y = 2$, then $z = 9$, which is not prime.

10.

Since there is no "there exists infinitely many quantifier" we use the following trick. We show that for every integer x there exists a strictly larger integer y that is a perfect square.

So the statement is true and can be written as

$$\forall x \exists y(y > x \wedge \exists k(y = k^2))$$

Proof. For any integer x , take $y = (|x|+1)^2$ which is guaranteed to be a perfect square and strictly larger than x . $\square$

Exercise 3.

Decide for each of the following claims whether it is logically valid. If it is valid, briefly justify why. If it is false, provide a counterexample using the universe of integers $\mathbb{Z}$.

1. $\forall x(P(x) \vee Q(x)) \equiv (\forall x P(x) \vee \forall x Q(x))$
2. $\exists x(P(x) \wedge Q(x)) \models (\exists x P(x) \wedge \exists x Q(x))$
3. $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$
4. $\forall y \exists x P(x, y) \models \exists x \forall y P(x, y)$
5. $\star (\forall x P(x) \rightarrow \forall x Q(x)) \models \forall x(P(x) \rightarrow Q(x))$

Solution 3.

1.

The claim is false.

Counterexample: Let $P(x) = 1$ if x is even and $Q(x) = 1$ if x is odd. Then LHS is true, but RHS is false.

2.

The claim is true.

If $\exists x(P(x) \wedge Q(x))$ holds, there exists some c such that both $P(c)$ and $Q(c)$ are true, which directly implies $\exists x P(x) \wedge \exists x Q(x)$.

3.

The claim is true.

If there exists a fixed y_0 such that $P(x, y_0)$ holds for all x , then for any arbitrary x we can choose $y = y_0$ to satisfy $\exists y P(x, y)$.

4.

The claim is false.

Counterexample: Take $P(x, y)$ as $x > y$. For any y we can choose $x = y + 1$, but there is no single integer x greater than all integers y .

5.★

The claim is false.

Counterexample: Let $P(x) = 1$ if $x > 0$ and $Q(x) = 1$ if $x < 0$. LHS evaluates to true, but RHS is false for $x = 1$.

★ **Exercise 4.**

Let the universe U be the set of all students taking an exam.

Let the predicate Q be defined as

$$Q(x) = \begin{cases} 1, & \text{If student } x \text{ passes the exam} \\ 0, & \text{Else} \end{cases}$$

Describe the following formula in natural language and determine if it is a tautology.

$$\exists x(Q(x) \rightarrow \forall y Q(y))$$

Solution 4.

The formula can be described as

"There exists a student such that if that student passes the exam, then every student passes the exam."

The formula is a tautology.

Proof. There are two cases. Either every student passes the exam or there exists at least one that fails.

Case 1: Every student passes the exam.

Since every student passes $\forall y Q(y)$ is true. Therefore, $Q(x) \rightarrow \forall y Q(y)$ is true regardless of $Q(x)$. Hence, such a student x exists.

Case 2: At least one student fails the exam.

Let χ be a student that fails. Since $Q(\chi)$ is false, the implication is vacuously true. Therefore χ , satisfies the formula.

Since for any exam session either Case 1 or Case 2 must hold, the formula is always true and therefore a tautology. $\square$

Note: This formula is called Drinker Paradox. You can find more on it here https://en.wikipedia.org/wiki/Drinker_paradox